

Shortcuts

Ctrl Shift A : Calculate Selected Area

Ctrl Shift G : Graph Historical Data

Ctrl Shift H : Historical Data Function

Ctrl Shift I : Indicative Data

Ctrl Shift P : Price/Yield Function

Ctrl Shift S : Calculate Selected Area

Ctrl Shift W : Calculate Workbook

Click on an Add-In function

Ctrl Shift C : Display Data Summary

Ctrl Shift D : Display Function Builder

Ctrl Shift E : Display Error Message

Ctrl Shift R : Display PDF Report or Collateral Analysis Report

CMO Abbreviations

Issuer	Example
FNMA w/(tranche)	fnma11.118(nb)
FHLMC w/(tranche)	f3250(nc)
GNMA w/(tranche)	g06.62(ko)
FHLMC Strip w/(tranche)	fhl.pc.244(po)
FNMA Strip w/(tranche)	fn.tr.375(l)
FNMA 2011-49 Tranche #2	fnma11.49(#2)
Bloomberg Ticker	GNR 11-139 MA

Mortgage Generics

TickerCoupon followed by “-Gen”

Ticker	Product
GNMA	30yr GNMA
GNMAII	30yr GNMA II
FNMA	30yr FNMA
FHLG	30yr FHLMC Gold
FNCT	20yr FNMA
FNCI	15yr FNMA
FHNG	15yr FHLMC Gold
GNMG	15yr GNMA
GNMGII	15yr GNMA II
FGTW	20yr FHLMC Gold

Example	Description
FNMA4-Gen	Generic 30yr FNMA
FNMA6-2010-PROD	FNMA 6% 30yr 2010 PROD
FNMA5.5-2008-LLB	FNMA 5.5% 2008 Low Loan Balance
FNMA4.5-2009-MLB	FNMA-4.5% 2009 Medium Loan Balance
FHLG5-2010-HLB	FHLMC Gold 5% 2010 High Loan Balance

Mortgage Pool

Description	Ticker	Example
FNMA Pool	n + Pool ID	n902558
GNMA Pool	g + Pool ID + x or m	g3921m
FHLMC Pool	f + Pool ID	f847773
Bloomberg Ticker		FN 902768

Utility Functions

Yield Book > Utility Functions

Workday Function – Similar to Excel's Workday function, but has a built in holiday calendar for major markets. Additionally, it removes the need for Excel's volatile 'Today' function.

Solver – Solves both Yield Book and non-Yield Book functions that require an iterative approach. It is similar to Excel's Goal Seek, but in a more powerful function form.

Tick/Decimal Converter – Converts decimals to ticks and vice versa. Applies mathematical operations to ticks.

Data Table – Used to output the Yield Book data stored in a particular cell. The data is output in text format (as a value).

Cash Flow Data Table – Used to output the Yield Book cash flow data stored in a particular cell or to a text file. The data is output as text in a non-function form.

Historical Data/Time Series

Historical data is available at a single point in time or in a time series function. Users can display data for a single date, or embed data from a defined range of dates in a single cell. Time series can be analyzed and graphed using Yield Book Add-In functionality.

Yield Book > Historical Data > Historical Data Function

Govt Curves	BMR	ABS
Securities	Indexes	CDS
Exch Rates	CMBS	TBA
Benchmarks	Swaps/Vols	

Collateral Analyzer

Yield Book > Additional MBS Analysis > Collateral Analysis

Allows the user to access interactive collateral analysis on Agency and Non-Agency CMOs.

Access the function via the drop down menu (YieldBook > Additional MBS Analysis>Collateral Analysis) or by hitting Ctrl+Shift+R on an Indicative Data function.

Auto P/Y Logic

By Price		
Price	Values > 25. To enter a price, append a “d”	90 or 12d
32 nd Price	Separate price and # of 32nds by a dash. To add one 64 th to this, add a “+”	90-20,90-20+
Full Price	Full price followed by “q”	99q
Unadjusted Price for Index-Linked Notes	Unadjusted price followed by “up”	105up
By Yield		
Yield	Values <= 25. To enter a yield greater than 25, append a “y”	10 or 27y
Discount Yield – Bills	Yield value followed by a “ds”	7ds
CD Yield	Yield value followed by a “cd”	5cd
Yield to First Call or Put	Append a “c” or “p” to a yield	8c or 8p
Yield to Worst Call	Yield value followed by a “w”	8w
Simple Yield	Value followed by “sy”	4sy
By Spread		
OAS	OAS in basis points followed by “o”	55o
Spread to Eurodollar Curve	Spread in bps followed by “e”	20e
Asset Swap Spread – Corporate	Spread in bps followed by “as”	25as
Spread to Actual Govt Curve	Spread in bps followed by “/i” (“I” for interpolated point on the govt curve using actual years)	120/i
Spread to Nominal Govt Curve	Spread in bps followed by “/j” (“J” for interpolated point on the govt curve using nominal maturity years for the on-the-run treasuries regardless of auction cycle)	120/j
Discount Margin (BEQ Effective Margin) and To Call	Discount margin followed by “m”, or “mc” for discount margin to call	154m or 145mc
Spread to Govt Rate	Spread in bps followed by “/” and benchmark year	120/30
Spread to Call	Spread in bps followed by “c”, “/” and benchmark year	120c/10
Spread to Put	Spread in bps followed by “p”, “/” and benchmark year	120p/10
Spread to Actual Govt Bond	Spread in bps followed by “/act” or “/” and a specific treasury (input as TickerCoupon, Maturity, or the CUSIP)	50/act or /us7.625,25
Spread to Swap Curve	Spread in bps followed by “/n” (50 bps over the interpolated point on the swap curve)	50/n
Spread to Swap Rate	Spread in bps followed by “/5n” (50 bps over the 5yr swap rate)	50/5n
BMA OAS (European Callable/Non-Callable)	BMA OAS for agencies in bps followed by “a”	100a
Zero Volatility Spread to Pricing Curve	Spread in bps followed by “/z”	20/z
Zero Volatility Spread to Treasury Curve (ARM)	Spread in bps followed by “/zi”	50/zi
Zero Volatility Spread to Swap Curve (CMO, ARM, Mortgage Pool)	Spread in bps followed by “/zn”	50zn
OAS Used to Create Par Bond and Solve for Coupon	Spread in bps followed by “cc”	40cc
Flat Spread Curve for CDS	Spread in bps followed by “flat”	50flat
Priced Spread at TBA OAS	Spread in points followed by “ps” (“ps” for point price spread)	0.5ps
Other		
Effective Reset and To Call	Effective reset followed by “er”, or effective reset to call followed by “xc”	150er or 135xc
Volatility for Derivatives	Value followed by “v” (“lv” for long rate volatility for swaptions)	25v
Volatility and Underlying Price for Options	Price volatility followed by “~” and price for underlying bond	20-100
Single Vol for Corporates	Pricing level followed by “@”, volatility value in %, “vol”	95@6vol
Real Yield Beta for TIPS	Pricing level followed by “@”, beta value, “beta”	99@0.5beta

Portfolio Definition

Yield Book > Portfolio Analysis > Define Portfolio

Security ID

Par Amount

Optional User Data

...

...

...

Portfolio Data

The Add-In assumes the first two columns correspond to Security ID and Par Amount. Portfolio Data and Portfolio Data Column Names must be defined in Portfolio Definition.

Security ID	Par Amount	Level	Client
912810pu6	100	100.1	Client A
31396pzn4	150	50/n	Client B
912828g23	200	50	Client B
FNMA6.5-gen	100	101	Client A

Syntax to reference Optional User Data: u.ColumnName
Example: u.Level, u.Client

Sector Files

Under Portfolio Analysis, users create a sector file by defining Sector File Data using the following syntax: Number, Name, Condition.

Number

Name

Condition

...

...

...

Sector File Data

For conditions, choose a keyword and follow it with an operator (=,<,>,etc.) and value. "And"/"/"or" can be used to combine conditions.

Number	Name	Condition
1	WAL 0-3	WAL<=3
2	FNMA 0-3	WAL<=3 and Ticker='FNMA'
3	GNMA 0-3	(Ticker='GNMA' or Ticker='GNMAII') and WAL<=3

Once created, the sector file (i.e. **YBSECTR_WAL**) can be referenced in the **Data Table Writer** under **Output Options** or **Portfolio Summary** under **Portfolio Analysis**

Visual Basic in the Add-In

The Add-In can be automated using Visual Basic. Please see the guide on our website ([www.yieldbook.com > Support > Yield Book Add-In](#)) for more information.

Scenario Analysis

Yield Book > Scenario Rate of Return

- **Bond and Pricing Data** - Similar inputs to the PY function
- **Horizon Pricing** - Defines the method and value at which you wish to price the bond on the horizon date
- **Scenario** - Defines the horizon period and yield curve shifts, as well as the timing of those shifts

Index as a Portfolio

Yield Book > Portfolio Analysis > Define Index As Portfolio

- Allows users to conveniently define an index as a portfolio
- User selects Index, As of Date, and Price Date
- As of Date and Price Date can be referenced from a cell
- Combine multiple indexes together

Optimization

Yield Book > Portfolio Analysis > Portfolio Optimization

Objective/Constraint

Weight

Total/Average

Sector

Side

OAS=Max

Objective

BaseMarketValue = 100000

EffectiveDuration = BL

BaseMarketValue = BLD

Market

Total Average

AssetClassCode='TG'

Candidate

Constraint(s)

Objective/Constraints

- Use Max or Min for Objective
- Choose a keyword and follow it with an operator (=,<,>,etc.) and value
 - BL = Base Line (Absolute term)
 - BLD = Base Line Distribution (Percentage term)

Weight – On the portfolio level, select par or total market as the averaging keyword. Leave blank for default weight.

Total/Average – On the portfolio level, the keyword can be totaled (i.e. Market Value) or averaged (i.e. Duration).

Sector – Apply the constraint to the entire portfolio or a specific sector.

Side - "Candidate" - Applies to candidate bonds only; "Starting" - Applies to the starting portfolio only; Left Blank - Applies to the final solution

Prepayment Options

Prepayment Speed	Description
PSA	Use a constant PSA speed
CPR	Use a constant CPR rate
Model (RMBS and CMBS)	Use the default Citi Prepayment Model for either RMBS or CMBS
CPB (CPR to Balloon)	Use a constant CPR until the security's Balloon Date
PPV (Prospectus Prepayment Vector)	Use prepayment vectors specified by the security's prospectus
MTB (Use Model Until Balloon Date)	Use the Citi Prepayment Model until the security's Balloon Date
ABS	Use the ABS prepayment model
CPY	Use a constant CPR during a security's yield maintenance period
HEP (Home Equity Loans ABS)	Use Citi's Home Equity Loan Prepayment Model
MHP (Manufactured Housing Prepayment)	Use Citi's Manufactured Housing Prepayment Model
New Model	Use the newest version of the Citi Prepayment Model before it is moved to the "Model" selection
Old Model	Use the previous production version of the Citi Prepayment Model

User Bonds

Swap

Swaption

Corporate

CUSIP

SWAP1234

CUSIP

SWAPTION

CUSIP

CORPORATE

TCKR

SWAP

TCKR

SWAPTION

SECTYPE

BOND

MATDATE

12/12/2012

MATDATE

12/12/2022

COUP

6

SECTYPE

SWAP

SECTYPE

SWAPTION

MATDATE

01/01/2017

SECSUBTYPE

SWAP

SECSUBTYPE

SWAPTION

DTDATE

01/01/2007

SECSUBTYPE2

SWAP

SECSUBTYPE2

SWAPTION

FCPDATE

07/01/2007

RDMTYPE

STANDARD

CURRENCY

USD

PAYISSDATE

12/12/2007

NOTIFYTYPE

BUSINESS

PAYTYPE

FLOATING

SWPNNOTIFDAY

2

PAYFREQ

4

OPTAMEURO

EUR

PAYDAYCOUNT

Act/360

OPTIONTYPE

PAYFIXED

PAYCRNCY

USD

SWPNFSEXDT

12/12/2012

PAYINDEX

LIB3

SWTNFINALDT

12/12/2012

PAYRESET

ADV

EXERPRICE

100

RCVISSDATE

12/12/2007

SWPNEXFREQ

1

RCVTYPE

FIXED

COUP

5

RCVFRO

4

PAYFREQ

4

RCVDAYCOUNT

30/360

DAYTYPE

Act/Act

RCVCRNCY

USD

FLTINDEX

LIB3

RCVCoupon

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Yield Book > Define User Data > User Bond

- User selects keywords and data to define Range(s)
- The User Bond Manager is a central repository to store a user's Add-In User Bonds
- Go to [www.yieldbook.com](#), under "Documentation" to view additional examples

Futures

Type	Ticker Base	Description	CUSIP Base
US Treasury Futures	TU	Two Year Contract	912821
	3Y	Three Year Contract	913812
	FV	Five Year Contract	912822
	TY	Ten Year Contract	914996
	US	Long Bond Contract	912815
German Govt Futures	GBS	SCHALTZ Contract	29105Q
	GBM	BOBL Contract	29105P
	GBL	BUND Contract	29105N
	GBB	BUXL Contract	29105T
Gilt Futures	TYG	2-Year GILT Contract	38104C
	FYG	5-Year GILT Contract	38402P
	FLG	Long GILT Contract	38402Y
CHF Futures	FB	10-Year Swiss Contract	80999J
JPY Govt Futures	JB	10-Year MINI JBG Contract	87199X
	JGB	10-Year JGB Contract	871993
KRW Govt Futures	KBS	3-Year KTB Contract	KR003F
	KBM	5-Year KTB Contract	KR005F
	KBL	10-Year KTB Contract	KR010F
AUD Govt Futures	TYB	3-Year AUD Govt Contract	124912
	XYB	10-Year AUD Govt Contract	124914
CAD Govt Futures	CBG	10-Year CAD Govt Contract	134996
US Agency Futures	WJ	10-Year Agn Contract	914889

Type	Ticker Base	Availability	CUSIP Base
Euro Dollar Futures	ED	Available quarterly out 10 years	296996
Euro Yen Futures	EY	Available quarterly out 2 years	866995
Euribor Futures	EE	Available quarterly out 5 years	29106M
3mo CAD Futures	BAX	Available quarterly out 2 years	108998
Short Sterling Futures	EP	Available quarterly out 5 years	912800
FED FUND 30 Day FUT	FF	Available monthly out 3 years	451992
2YR OTR TSY FUT	TW	Available monthly out 1 month	923821
5 YR OTR TSY FUT	FI	Available monthly out 1 month	922966
10 YR OTR TSY FUT	TE	Available quarterly out 1 month	923822
5 YR INT RATE SWAP	NJ	Available quarterly out 6 months	323991
10 YR INT RATE SWAP	NI	Available quarterly out 6 months	322991
LIFFE 2Y EUR Swapnote	TWS	Available monthly out 1 month	LIFF2E
LIFFE 5Y EUR Swapnote	O	Available monthly out 1 month	LIFF5E
LIFFE 10Y EUR Swapnote	P	Available monthly out 1 month	LIFF0E
LIFFE 2Y USD Swapnote	USW	Available monthly out 1 month	LIFF2D
LIFFE 5Y USD Swapnote	USO	Available monthly out 1 month	LIFF5D
LIFFE 10Y USD Swapnote	USP	Available monthly out 1 month	LIFF0D

Examples

USM.Z

EDH6.Z

GBMM.Z

912815M1 June 2011 USD Long Bond Contract

296996H6 March 2016 Euro Dollar Contract

29105PM1 June 2011 GER BOBL Contract

Futures Ticker Methodology:
Base + Expiry Month + Expiry Year† + .Z*

† One digit number for Expiry Year, i.e. 1 for 2011, 5 for 2015

* Expiry Year only necessary when multiple contracts exist for expiry month, and it is not necessary for U.S Treasury Futures

Futures CUSIP Methodology:
CUSIP Base + Expiry Month + Expiry Year

Expiry Codes		
F January	G February	H March
J April	K May	M June
N July	Q August	U September
V October	X November	Z December